MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW EFU-5/A, FUEL TOTALIZING, 0-120,000 PPH

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specifications covers a remote-indicating-summation fuel-flow indicator designated EFU-5/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-I-7566	Indicator System, Fuel Flowmeter, General Specification for
MIL-P-9107	Test Reports, Preparation Of
MIL-I-25907	Indicator, Rate of Flow EFU-4/A, Fuel Totalizing, 500-15,000 PPH

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Logistics Center/MMEDO, Tinker AFB OK 73145-5990, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-26514	Polyurethane Foam, Rigid or Flexible, For Packaging

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MS25043	Cap, Electrical Connector Receptacle, Dust
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.6.4, 6.2 and 6.3).

3.2 General specification. The requirements of MIL-I-7566 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 DESIGN AND CONSTRUCTION

3.3.1 Dual pointers. The summation flow-rate indicator shall be designed with dual pointers to indicate a rate-of-flow from 0 to 120,000 pph. The fast pointer shall indicate 10,000 pph per revolution and the slow pointer 120,000 pph in 1.2 revolutions.

3.3.1.1 Voltage. The indicator shall operate from an interrupted 400-cycle voltage of 1V per 1000 pph flow rate supplied by 2 or more EFU-4/A rate-of-flow indicators in accordance with MIL-I-25907.

3.3.1.2 Components and pointers. The indicating section of the indicator shall contain a low inertia 2-phase motor, other components, and pointers to indicate the rate of flow.

3.3.1.3 Amplifier. An amplifier shall be incorporated as an integral unit, contained within the indicator case.

3.3.1.4 Indicator rate. The indicator shall be designed to indicate a change of flow rate from 0 to 120,000 pph in not more than 20 and not less than 10 seconds at room temperature.

3.4 Case. The case shall conform to figure 1 and shall be sealed hermetically.

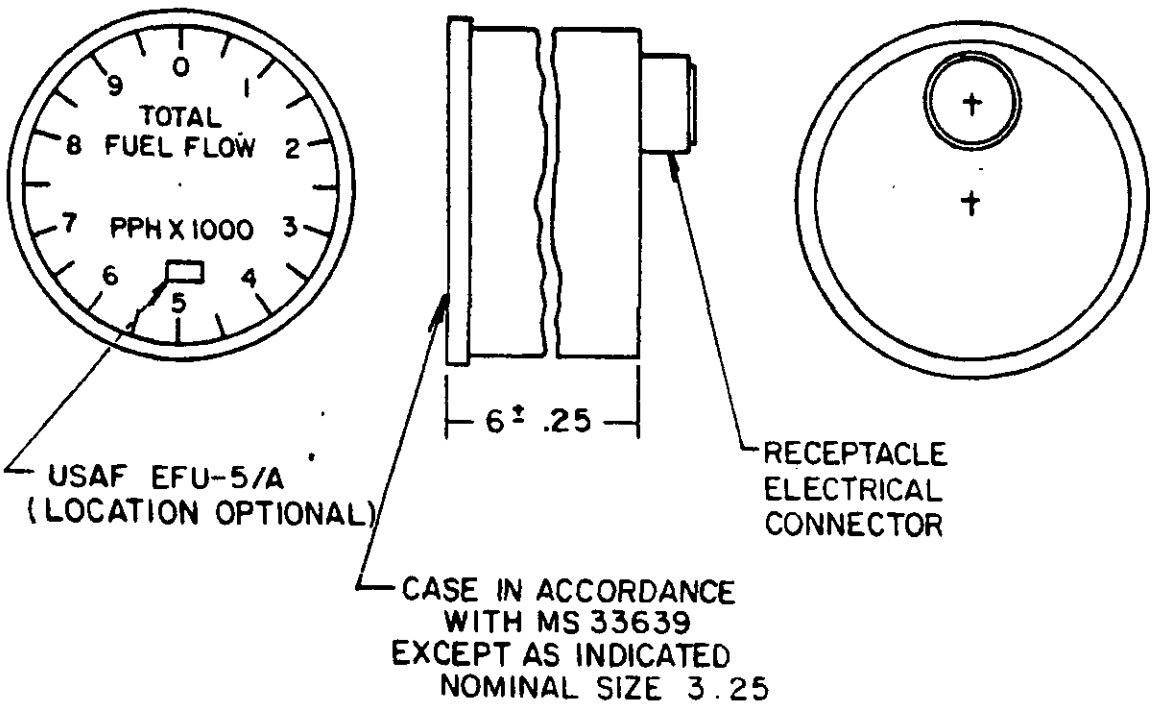
3.5 Dial

3.5.1 Minimum diameter of the dial. The minimum diameter of the dial, as measured across the outside ends of the graduations, shall be 2-29/32 inches. The dial for the summation flow-rate indicator shall conform to figure 1. The electrical connector shall be located on the vertical center line with the connector guide key at the top. When mated, the mating connector dimension shall not extend above the case diameter.

3.5.2 Dial-to-cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.187 inch.

3.6 Pointers. The short pointers shall conform to MS33585-4 and the long pointers to MS33585-3 except that the taper of the tip shall be 5/16 inch.

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DIMENSIONS IN INCHES

FIGURE 1. Dial and case.

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3.7 Power input. The power supply shall be 115 ±5V 400 ±20 cycles single phase. The waveform of the supply voltage may be nonsinusidal with a crest factor of 1.414 ±10 percent and with harmonics whose respective amplitudes are less than 5 percent of the fundamental. No damage shall result to the instrument if a voltage or frequency of 10 percent exceeding the maximum or minimum specified is applied.

3.8 Cable length. Accuracy and sensitivity of the indicator system shall not be adversely affected by varying the length of interconnecting cable between the rate-of-flow indicators and the total rate-of-flow indicator for a cable capacitance up to 1200 uuf.

3.9 Cruise control output. The indicator shall contain a potentiometer whose slider is positioned proportional to indication. The potentionmeter is a 12-turn, pot-rated at 2.4 turns for 10,000 ohms. The acceptable tolerance is ±41 ohms between 8331 to 16,666 ohms and ±104 ohms elsewhere. The total rating is 50K. The potentiometer terminals shall be brought directly to the connector with no other internal connection. External connections to this potentiometer shall cause no more than 1 watt of power to be dissipated inside the instrument.

3.10 Weight. The weight of the indicator and self-contained amplifier shall not exceed 2.5 pounds.

3.11 Electrical receptacles. The indicator shall be provided with an electrical receptacle in accordance with MS33678-18-1PW except that it shall be designed for hermetic sealing.

3.11.1 Electrical connector. The electrical connector receptacle shall be equipped with an MS25043-18 cap and gasket to protect the electrical contact surfaces from water vapors during shipment and storage.

3.12 MARKINGS

3.12.1 Fluorescent luminescent. The following marking shall be finished in fluorescent-luminescent material. The markings shall conform to MS33558 and have the following dimensions:

	Height or Length Inch ±.016	Width of Line or Graduation Inch ±0.005
Scale numerals 1, 2, 3, etc	0.219	0.031
Graduations opposite numerals	0.219	0.031
500-pound graduations	0.156	0.020
100-pound graduations	0.094	0.016
Lettering PPH x 1000	0.125	0.020
Lettering 10,000 PPH	0.062	0.016
Lettering TOTAL FUEL FLOW	0.125	0.020

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3.12.2 Durable dull black. The following markings shall be marked permanently and legibly on the dial. These and all other markings not otherwise specified shall be furnished in durable dull black. The height of the markings shall be as follows:

	Height Inch
Lettering USAF and EFU-5/A	0.047

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification as specified in MIL-I-7566.

3.14 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 First article designation. For the purpose of this specification, the qualification tests specified in MIL-I-7566 shall be identified as first articles tests.

4.2 General specification. Sampling, inspection, and test procedures shall be in accordance with MIL-I-7566 and as specified herein.

4.3 TEST EQUIPMENT STANDARDS

4.3.1 Master transmitter. Indicators shall be tested with a master transmitter as specified in MIL-I-7566 except that for system indication, the master transmitter shall provide an output equivalent to the summation output signal of 2 to 8 individual rate-of-flow indicators.

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4.4 Performance. The performance of the indicator shall be checked against tables I, II, and III and shall not exceed the tolerances specified therein.

TABLE I.

Indicator Scale Error.

Flow Rate PPH	Room Temp Tol ±PPH	Low Temp Tol +PPH	Low Temp Tol -PPH	High Temp Tol ±PPH
5,000	100	100	240	125
10,000	100	100	280	125
25,000	100	125	300	185
30,000	150	150	320	225
40,000	200	200	380	350
50,000	240	240	440	450
60,000	280	280	500	550
70,000	320	320	560	650
80,000	360	360	620	750
90,000	400	400	680	850
100,000	440	440	740	950
120,000	520	520	860	1050

4.4.1 Endurance. The indicator shall be tested as indicated in MIL-I-7566 except that the total cycles shall be 25,000.

TABLE II.

Indicator Friction and Vibration Error.

Test	Tolerance PPG
Position error	100
Vibration	
Pointer oscillation, double amplitude	50
Pointer variation	50
Vibration failure	
Scale error	Room temp limits of table I
Friction error	50

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TABLE II. (Cont)

Indicator Friction and Vibration Error.

Test	Tolerance PPH
Friction	
Room temperature	50
High temperature	50
Low temperature	100
Response time	20 sec max, 10 sec min
Power input	20 watts max
* Position error at 20,000 and at 70,000 PPH	

TABLE III.

Voltage and Frequency Variation.

Flow Rates	Tolerances ±PPH
5,000-25,000	150
30,000	200
40,000-120,000	300

4.4.2 Sensitivity. The indicator shall be connected to the master transmitter in a manner to provide an indication. A change in the transmitter setting equivalent to 500 pph shall be made and, on the indicator, a change of 500 ±100 pph shall be shown in the ranges from 0-30,000 and a change of 500 ±150 pph shall be shown in the range from 30,000 to 100,000. This test may be conducted at any indication at the discretion of the contracting activity.

4.4.3 Cruise-control output error. The resistance of cruise-control output shall be in the same proportion of the total potentiometer resistance as the scale reading is of full scale to within the following limits. For a full-scale value of 120,000 pph the deviation from scale error shall be less than ±100 pph in the range from 20,000 to 40,000 pph and ±250 pph above and below this range.

4.4.4 INDIVIDUAL TESTS

4.4.4.1 Systems test. A room-temperature system test shall be conducted at indicated flow rates, utilizing summation output signals of 8 individual indicators. A master transmitter or other suitable means, having the same electrical characteristics as indicators in accordance with MIL-I-25907, shall be used to actuate the individual indicators. The indicated summation flow rate shall equal the sum of the indicated individual flow rate within ±2 percent tolerance at the 16,000, 24,000, 48,000 and 72,000 pph test points.

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4.5 SAMPLING PLAN A

4.5.1 Fogging. The indicator shall be energized and placed in a $+71^{\circ} \pm 2^{\circ}C$ controlled ambient for a minimum of 1 hour. Upon completion of this period and while still at this temperature, an ice cube shall be rubbed on the indicator glass face for 1 to 2 minutes. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog. Evidence of fogging shall be cause for rejection. This test may be conducted in conjunction with the high temperature test.

4.6 First articles inspection. The supplier shall subject three indicators to the first article inspection specified in 4.6.3.

4.6.1 Disposition of first article test samples. When first article samples are required, the supplier shall submit the samples to the contracting activity for:

a. Review of the mechanical construction.

b. Performance of any specification tests deemed necessary after review of the supplier's test report.

4.6.2 First article tests. First article tests shall consist of all the test specified in MIL-I-7566 and herein.

4.6.3 Inspection of packaging. Tests of methods of preservation and packaging shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification.

5. Packaging

5.1 Preservation. Preservation shall be level A, C, or industrial as specified (see 6.2)

5.1.1. Level A.

5.1.1.1. Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2. Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3. Preservation application. Not applicable.

5.1.1.4. Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

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5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2. Level B. Indicator packaged as specified in 5.1.1. shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords protection from domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The EFU-5/A indicator covered by this specification is intended for use to indicate the total fuel flow rate indicated for multiengine aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Number of first article samples and point of testing.

c. Level of packaging and packing desired.

6.3 First article. A first article is required for inspection and test (see 3.1 and 4.6.4).

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6.4 Subject term (key word) listing.
Fluorescent-Luminescent
Fuel
Indicator
Multiengine-Aircraft
Range 0-120,000 PPH
Rate of Flow
Totalizing

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE -99

Preparing activity:
AIR FORCE 71

Project No:
6620-F426

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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